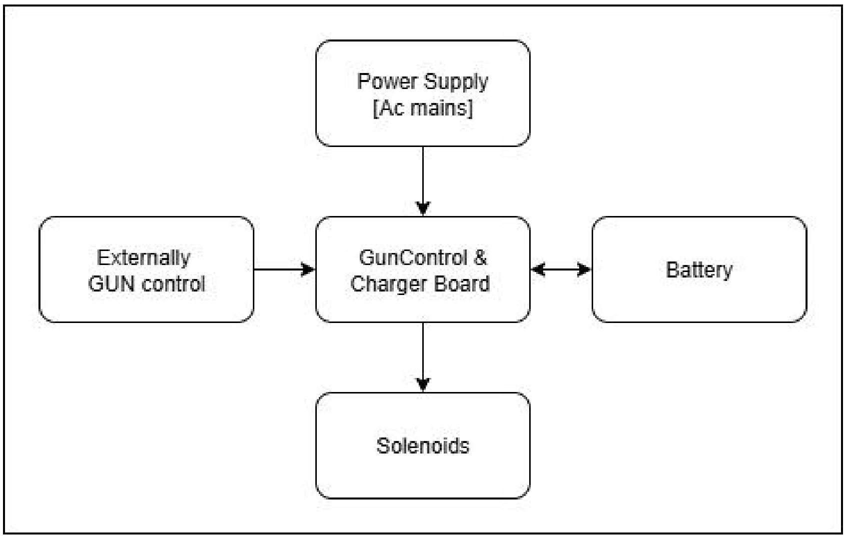
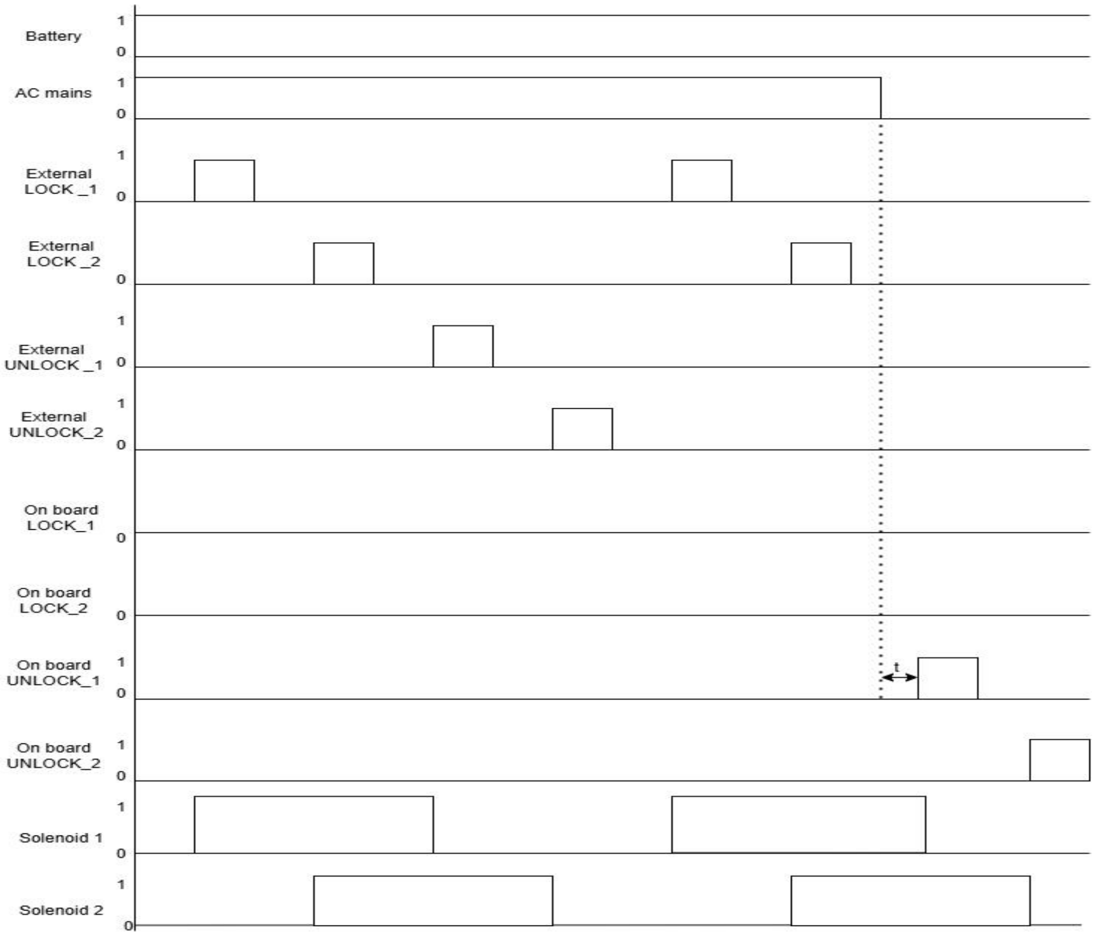
Battery charger and Gun Control



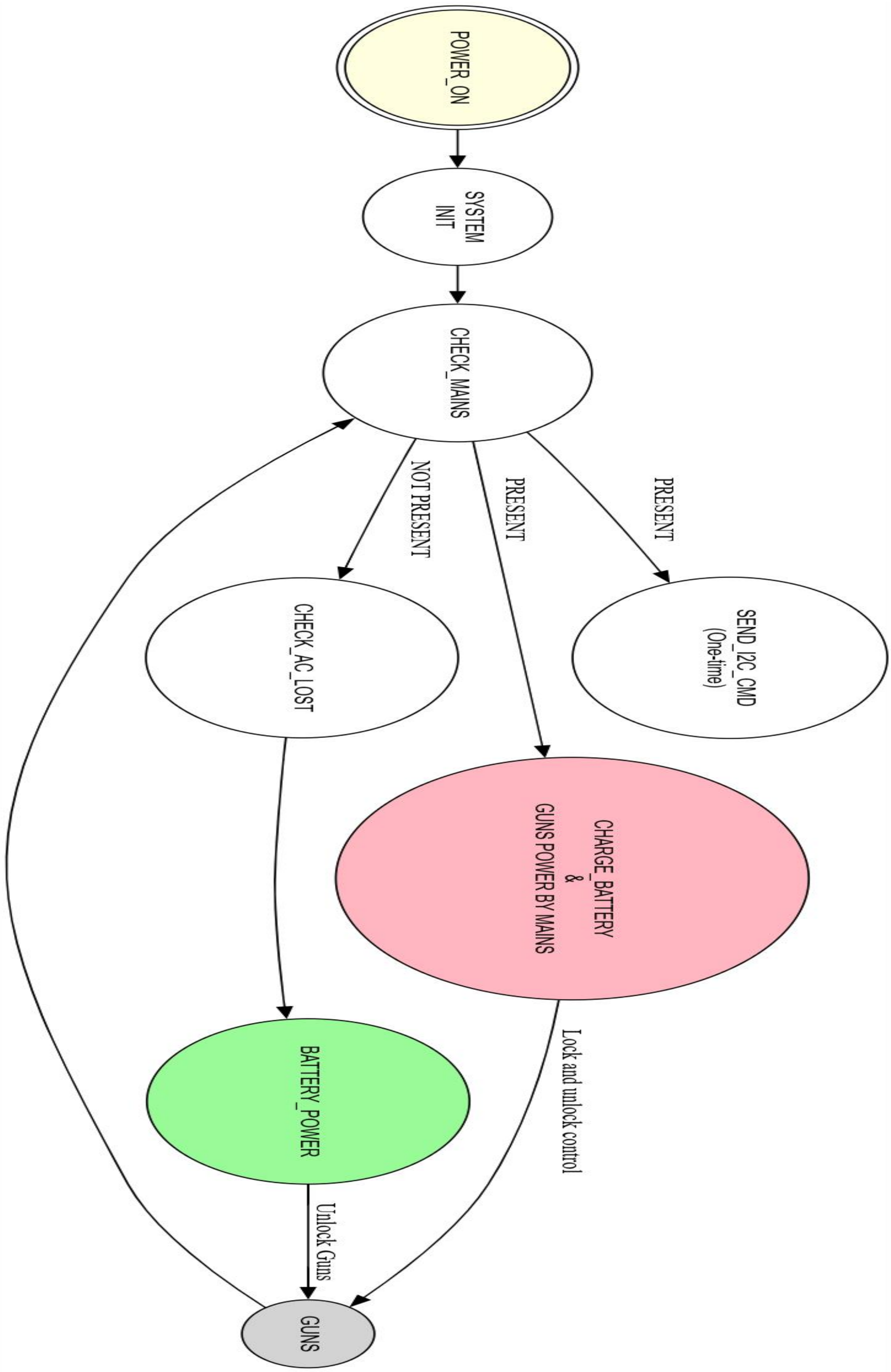
Blockdiagram



Note:

"'t' indicates that the controller has detected AC mains is OFF, and a 5-second delay is applied."

Waveforms



Finite state machine

Project Flow – System Operation Description

1. System Power-Up

* IF +12 V DC (AC mains) is present at startup
  → MCU initializes all peripherals (I^2C, GPIO, timers)
  → Sets AC mains status = PRESENT
  → Sends I^2C command to charger IC (BQ24735) to start charging the battery.

2. Normal Operation – AC Mains Available

* IF AC mains detect = PRESENT
  → Charger IC supplies system load and charges the battery.
  → MCU keeps all solenoid control pins in output, low state (idle).
  → Solenoid lock/unlock actions are driven externally.

3. AC Mains Loss Detection

* IF AC mains detect = NOT PRESENT
  → MCU waits 1 second for confirmation (noise/false trigger filter).

  * IF mains still absent after delay
    → MCU sends 100 ms unlock pulse to both solenoid H-Bridge inputs.
    → MCU resets control pins to idle state.
    → Updates state = AC LOSS MODE.

4. Battery Operation Mode

* IF system is in AC LOSS MODE
  → Battery powers MCU, H-Bridge drivers, and solenoids.
  → Charger IC is disabled to minimize drain.
  → Only essential operations are performed.

5. AC Mains Recovery

* IF AC mains detect = PRESENT again
  → MCU updates AC mains status = PRESENT
  → Sends I^2C command to charger IC to re-enable charging.
  → System transitions back to Normal Operation Mode.

6. H-Bridge & Solenoid Control

* IF unlock action is triggered (during AC loss or specific events)
  → MCU drives the corresponding H-Bridge with a precisely timed pulse.
  → Ensures reliable solenoid actuation and minimal power consumption.